RAY OPTICS AND OPTICAL INSTRUMENTS

Important formulas and concepts directly from the NCERT-

- Ray optics – the straight line path along which the light travel in a homogenous medium.

- **Angle of deviation –** angle by which a straight line deviates when it hits a surface and get reflected in same medium.

- **$\delta = \pi - 2i$ (i = angle of incidence)**

- **relation b/w velocity of object and image for plane mirrors- (**velocities perpendicular to mirror**)**

- **$\upsilon_i = -\upsilon_o + 2\upsilon_M$ (υi = velocity of image)**
  (υo = velocity of object)
  (υ_M = velocity of mirror)

- **spherical mirror –**

- **2 types – (1) concave = outer surface is polished (cave me light aaegi)**
  (2) convex = inner surface is polished

- **Mirror formula, $1/f = 1/u + 1/\upsilon$**

  - ✓ **$\upsilon = uf / (u-f)$**

- **focal length (f) for concave mirrors = negative**
  for convex mirror = positive

- **magnification, $m = -\upsilon/u = hi/ho$ (hi and ho =** heights of image and object**)**

  - ✓ **$m = f / (f-u)$**
  - ✓ **$m = (f-\upsilon) / f$**

- **power of mirror,**

$$P = 1/f$$

- **Refraction –** when light travels from one medium to another medium, it deviates from its path , this phenomenon is k/as 'refraction'.

- **refractive index(μ) –**

$$\mu = \text{velocity of light in vaccum / velocity of light in medium}$$
$$\mu = c / \upsilon$$

- $c = 1 / \sqrt{\mu_o \varepsilon_o}$
- $\upsilon = 1 / \sqrt{\mu_o \mu_r \varepsilon_o \varepsilon_r}$
- $\upsilon = \nu\lambda$ (λ = wavelength of light
  ν = frequency of light)
- $\mu_1 / \mu_2 = \upsilon_2 / \upsilon_1$
- ${}_1\mu_2 = \mu_2 / \mu_1$

- **Snell's law –** (for refraction)

$$\mu_i \sin i = \mu_r \sin r$$ (i = angle of incidence)
(r = angle of refraction)

- "When light travels from one medium to other medium, frequency remains same".
- $\mu_1 / \mu_2 = \lambda_2 / \lambda_1$

- **Shift through glass slab –**

$$\text{Shift} = t\,[1 - 1/\mu]$$

- **Real and apparent depth –**
  - Case 1 – observer in denser medium and object in rarer medium

$$d' = d / \mu$$

  - Case 2 – observer in rarer medium and object in denser medium

$$d' = \mu d$$

❖ **Critical angle (i_c)–**

$$\sin i_c = 1 / \mu$$

➢ **For cone ,**

$$R = h / \sqrt{\mu^2-1}$$

❖ **Prism –**

➢ Angle of prism, $A = r_1 + r_2$

➢ Minimum deviation, $\delta = i + e - A$

➢ **If one face silvered,**

$$A = r$$

➢ **Condition for minimum deviation,**

$$i = e$$
$$A = 2r$$

$$\mu = \sin(\delta_{min} + A/2) / \sin(A/2)$$

- **For thin prism,**

$$\delta_{min} = (\mu-1) A$$

❖ **Spherical refractive surface –**

$$\mu_2 / \upsilon - \mu_1 / u = (\mu_2-\mu_1) / R$$

❖ **Lens –**

(1) Convex lens –

1) **Inverted and diminished,**

$$m = \text{neg. and } |m| < 1$$

2) **Inverted and same size,**

$$m = \text{neg. and } |m| = 1$$

3) **Inverted and enlarge,**

$$m = \text{neg. and } |m| > 1$$

4) **Erect and enlarge,**

$$m = \text{positive and } |m| > 1$$

(2) Concave lens-

1) **Virtual, erect and diminished**

$$m = \text{positive and } |m| << 1$$

- ❖ **<u>Lens maker formula-</u>**

$$1/f = (\mu_l/\mu_m - 1)(1/R_1 - 1/R_2)$$

- ❖ **<u>Lens formula</u> –**

$$1/f = 1/\upsilon - 1/u$$

- ✓ $\upsilon = uf/(u + f)$
- ✓ $m = h_i/h_o = \upsilon/u$

- ➢ **focal length of convex lens = positive**
  concave lens = negative

SUMMARY

1. Reflection is governed by the equation $\angle i = \angle r'$ and refraction by the Snell's law, $\sin i/\sin r = n$, where the incident ray, reflected ray, refracted ray and normal lie in the same plane. Angles of incidence, reflection and refraction are i, r' and r, respectively.
2. The *critical angle of incidence* i_c for a ray incident from a denser to rarer medium, is that angle for which the angle of refraction is 90°. For $i > i_c$, total internal reflection occurs. Multiple internal reflections in diamond ($i_c \cong 24.4°$), totally reflecting prisms and mirage, are some examples of total internal reflection. Optical fibres consist of glass fibres coated with a thin layer of material of *lower* refractive index. Light incident at an angle at one end comes out at the other, after multiple internal reflections, even if the fibre is bent.
3. *Cartesian sign convention*: Distances measured in the same direction as the incident light are positive; those measured in the opposite direction are negative. All distances are measured from the pole/optic centre of the mirror/lens on the principal axis. The heights measured upwards above x-axis and normal to the principal axis of the mirror/lens are taken as positive. The heights measured downwards are taken as negative.
4. *Mirror equation*:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

where u and v are object and image distances, respectively and f is the focal length of the mirror. f is (approximately) half the radius of curvature R. f is negative for concave mirror; f is positive for a convex mirror.

5. For a prism of the angle A, of refractive index n_2 placed in a medium of refractive index n_1,

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\left[(A + D_m)/2\right]}{\sin(A/2)}$$

where D_m is the angle of minimum deviation.

6. *For refraction through a spherical interface* (from medium 1 to 2 of refractive index n_1 and n_2, respectively)

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Thin lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Lens maker's formula

$$\frac{1}{f} = \frac{(n_2 - n_1)}{n_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

R_1 and R_2 are the radii of curvature of the lens surfaces. f is positive for a converging lens; f is negative for a diverging lens. The power of a lens $P = 1/f$.

The SI unit for power of a lens is dioptre (D): 1 D = 1 m^{-1}.

If several thin lenses of focal length f_1, f_2, f_3,.. are in contact, the effective focal length of their combination, is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \ldots$$

The total power of a combination of several lenses is

$P = P_1 + P_2 + P_3 + \ldots$

7. *Dispersion* is the splitting of light into its constituent colour.

8. *Magnifying power m of a simple microscope* is given by $m = 1 + (D/f)$, where D = 25 cm is the least distance of distinct vision and f is the focal length of the convex lens. If the image is at infinity, $m = D/f$. For a compound microscope, the magnifying power is given by $m = m_e \times m_0$ where $m_e = 1 + (D/f_e)$, is the magnification due to the eyepiece and m_o is the magnification produced by the objective. *Approximately,*

$$m = \frac{L}{f_o} \times \frac{D}{f_e}$$

where f_o and f_e are the focal lengths of the objective and eyepiece, respectively, and L is the distance between their focal points.

9. *Magnifying power m of a telescope* is the ratio of the angle β subtended at the eye by the image to the angle α subtended at the eye by the object.

$$m = \frac{\beta}{\alpha} = \frac{f_o}{f_e}$$

where f_0 and f_e are the focal lengths of the objective and eyepiece, respectively.

POINTS TO PONDER

1. The laws of reflection and refraction are true for all surfaces and pairs of media at the point of the incidence.
2. The real image of an object placed between f and $2f$ from a convex lens can be seen on a screen placed at the image location. If the screen is removed, is the image still there? This question puzzles many, because it is difficult to reconcile ourselves with an image suspended in air without a screen. But the image does exist. Rays from a given point on the object are converging to an image point in space and diverging away. The screen simply diffuses these rays, some of which reach our eye and we see the image. This can be seen by the images formed in air during a laser show.
3. Image formation needs regular reflection/refraction. In principle, all rays from a given point should reach the same image point. This is why you do not see your image by an irregular reflecting object, say the page of a book.
4. Thick lenses give coloured images due to dispersion. The variety in colour of objects we see around us is due to the constituent colours of the light incident on them. A monochromatic light may produce an entirely different perception about the colours on an object as seen in white light.
5. For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification because we can keep the small object much closer to the eye than 25 cm and hence have it subtend a large angle. The image is at 25 cm which we can see. Without the microscope, you would need to keep the small object at 25 cm which would subtend a very small angle.